

	1.5	A1
6(a)	electric field perpendicular to magnetic field velocity perpendicular to both electric and magnetic fields $\left. \begin{aligned} F_E &= F_B \\ qE &= Bqv \\ v &= \frac{E}{B} \end{aligned} \right\}$	B1 B1 B1
6(b)(i)	magnetic force provides centripetal force $Bqv = \frac{mv^2}{r}$ $\frac{q}{m} = \frac{v}{rB}$ $\frac{\left(\frac{q}{m}\right)_P}{\left(\frac{q}{m}\right)_Q} = \frac{v_P}{v_Q} \frac{B_Q}{B_P} \frac{r_Q}{r_P} = \frac{r_Q}{r_P}$ $= 2$	C1 A1

6(b)(ii)					A1
	Particle	Mass	Charge	Charge-Mass Ratio	
	Protium	1 u	+ e	e/u	
	Tritium	3 u	+ e	$e/3u$	
	Alpha particle	4 u	+ 2 e	$e/2u$	
	P: protium Q: alpha				

6(b)(iii)	time to collision with wall is half period magnetic force provides centripetal force $\left. \begin{aligned} Bqv &= mr\omega^2 \\ Bq(r\omega) &= mr\omega^2 \\ Bq &= m\omega \\ \omega &= \frac{q}{m}B = \frac{2\pi}{T} \end{aligned} \right\}$ $\frac{T_Q}{T_P} = \frac{\omega_P}{\omega_Q} = \frac{\left(\frac{q}{m}\right)_P}{\left(\frac{q}{m}\right)_Q} = 2$ OR since both particles travel at the same v and Q travels twice the distance of P in one period, $\frac{T_Q}{T_P} = 2$	M1 A1 (M1) (A1)
7(a)(i)	An object will continue to be in a state of rest or move with constant	P1